

NIRANJANA RAM

Climate Risk & Geospatial Analyst | Coastal Hazards, Disaster Risk & Remote Sensing | Erasmus Mundus MSc

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PROFILE

Early-career climate and disaster-risk researcher with a physics foundation and an Erasmus Mundus MSc in Coastal Hazards, Risks and Climate Change Adaptation, and two years of applied research experience across India and Europe. Combines a strong GIS and remote-sensing foundation with quantitative risk modelling in Python (CLIMADA, uncertainty analysis) and coastal vulnerability index construction. Skilled in geospatial analysis, large-dataset processing on high-performance computing, and scientific writing.

TECHNICAL SKILLS

GIS & Remote Sensing: ArcGIS · QGIS · Google Earth Engine · remote sensing · spatial analysis · cartography

Programming & Databases: Python (GeoPandas, rasterio, pandas, numpy) · MATLAB · SQL (PostgreSQL / PostGIS)

Risk & Impact Modelling: CLIMADA · coastal vulnerability index construction · hazard–exposure–vulnerability assessment · uncertainty & sensitivity analysis (Monte-Carlo, Sobol) · high-performance computing

Numerical Coastal Modelling: Delft3D · SWAN · XBeach

Field & Data: GPS field data collection (Collector for ArcGIS) · geodatabase design · multi-format spatial data (GeoTIFF, shapefile, KML, GeoJSON)

Research: Scientific and technical writing · literature review

PROFESSIONAL EXPERIENCE

Research Intern (MSc Thesis Project) — IHCantabria, Universidad de Cantabria

Jan – Jul 2026

Environmental Hydraulics Institute of Cantabria, Santander, Spain

- Developed a probabilistic risk-assessment framework quantifying population-displacement risk from tropical-cyclone winds across the North Atlantic and Caribbean, conditioned on ENSO phases (El Niño / La Niña / Neutral).
- Developed income-based displacement impact functions linking cyclone wind speed to the fraction of population displaced, calibrated across income groups from observed records (IDMC), and integrated these with high-resolution population exposure across 500+ administrative units and ~460 million people in 33 countries.
- Implemented the full analysis in Python (GeoPandas, CLIMADA) with frequency-bias correction and Monte-Carlo-based uncertainty/sensitivity analysis (Sobol); ran large-scale simulations on the MareNostrum 5 high-performance computing cluster.
- Produced publication-quality maps and a written thesis with reproducible, documented workflows, within an international, interdisciplinary research team.

Research Assistant — KSCSTE-CWRDM

Oct 2022 – Oct 2023

Centre for Water Resources Development and Management, Kozhikode, Kerala, India

- Developed a composite Coastal Vulnerability Index (CVI) for the entire Kerala coast to assess vulnerability to sea-level rise — combining socio-economic, physical and environmental sub-indices — on the ICSSR-funded project under Dr. K.Ch.V. Naga Kumar.
- Conducted geospatial analysis (ArcGIS, QGIS — overlay, proximity, surface and multi-criteria analysis) and remote-sensing classification (NDVI), including multi-temporal coastal/shoreline change analysis.
- Designed and managed geodatabases and multi-format spatial data; automated workflows in Python and queried databases with SQL.
- Collected field data (GPS, Collector for ArcGIS); produced cartographic vulnerability maps and contributed to technical reports.

Disaster Management Team Member — Pragma

Mar – Jun 2024

Disaster Management & Community Resilience Programme, Gurugram, India

- Supported field-level disaster-preparedness and community-resilience activities with remote, vulnerable communities, as part of Pragma's grassroots disaster risk reduction work.
- Assisted in hazard and vulnerability mapping and stakeholder consultations with communities and local authorities.
- Contributed to data collection, report writing, documentation and outreach materials for disaster-preparedness initiatives.

EDUCATION

MSc Coastal Hazards, Risks, Climate Change Impacts and Adaptation (COASTHazar) — Erasmus Mundus Joint Master's

Sept 2024 – Sept 2026

Universidad de Cantabria (ES) · IHE Delft (NL) · Universidade do Algarve (PT) — EQF Level 7

Key subjects: coastal processes and morphodynamics; coastal hazards, risk theory and applied statistics; numerical and statistical coastal modelling (Delft3D); climate change impacts and adaptation; coastal protection and nature-based solutions; GIS and hydrographic surveying; AI and machine learning for coastal resilience.

Thesis: “Population Displacement Risk Assessment for Tropical Cyclone Winds in the North Atlantic under ENSO” — Python, CLIMADA, uncertainty analysis, HPC.

MSc Disaster Management

Aug 2020 – May 2022

Pondicherry University (Andaman Campus), Port Blair, India — 80% · EQF Level 7

Key subjects: disaster risk reduction and management; natural hazards; remote sensing (optical/thermal/SAR, LIDAR, Google Earth Engine); GIS and spatial analysis; geomorphology; geophysical exploration; climate change; preparedness, response and recovery.

Dissertation: “Impact of Very Severe Cyclonic Storms Hudhud and Vardah on Shoreline and Water Body Changes.”

BSc Physics

Aug 2017 – Apr 2020

Government College Madappally, University of Calicut, India — 80% · EQF Level 6

Key subjects: mechanics; electricity, magnetism and electrodynamics; thermal and statistical physics; optics; modern and quantum physics; solid-state and nuclear physics; electronics (incl. Intel 8085 microprocessor); mathematical and computational physics.

Professional Certificate — Advanced Geospatial Engineering Program (GIS)

Nov 2023 – Apr 2024

Avakaza Geoscience Research Technologies (AGSRT), Bengaluru, India

Key subjects: Python GIS automation (GeoPandas, ArcPy, rasterio); PostgreSQL/PostGIS and Spatial SQL; FME (ETL); GeoServer; Google Earth Engine (geemap); web GIS and dashboards; remote sensing.

LANGUAGES

English — full professional proficiency · Malayalam — native · Hindi — professional · Spanish — elementary